

Arrhenius equation (HL)

Higher level (HL)

Learning outcomes

By the end of this section you should be able to:

- (R2.2.12) Describe the qualitative relationship between temperature and the rate constant.
- (R2.2.12) Analyse graphical representations of the Arrhenius equation including its linear form.
- (R2.2.13) Determine the activation energy and Arrhenius factor from experimental data.

In section R2.2.9 (<https://app.kognity.com/study/app/preview-p/sid-426-cid-753301/book/the-rate-expression-and-its-experimental-id-45511/>) you learned that the rate of reaction depends on two things, the rate constant k and the concentration of the reactants raised to a power, represented by rate = $k[A][B]$.

You also learned about k in the previous section and how it is linked to the order of reaction and the rate of reaction. What exactly is k and what determines its value?

It is proportional to rate, so a higher value of k will mean a faster rate of reaction. Some reactions progress very quickly others are slow, is this because of k ?

The temperature dependence of the rate constant k

You have previously learned that increasing the temperature increases the rate of reaction (section R2.2.2–3a (<https://app.kognity.com/study/app/preview-p/sid-426-cid-753301/book/factors-affecting-the-rate-of-reaction-id-44714/>)). For example, milk will sour more quickly at higher temperatures than at lower ones. Since increasing temperature does not change the concentration of the reactants, it follows that the rate constant k is dependent on the temperature.

Generally speaking, the rate constant, k depends on two factors:

- The temperature — recall from the Maxwell-Boltzmann curves (section R2.2.4–5 (<https://app.kognity.com/study/app/preview-p/sid-426-cid-753301/book/maxwell-boltzmann-distribution-curves-id-44782/>)) that at lower temperatures the number of collisions that have energy equal to or greater than activation energy are small. If you increase the temperature the number of collisions with energy equal to or greater than the activation energy grows exponentially. This means increasing the temperature will increase the rate constant k .
- The presence of a catalyst — this means an alternative route with lower activation energy. This increases the rate of reaction, hence increases k . The rate constant, k , is therefore also dependent on the value of activation energy.

It was Svante Arrhenius who consolidated these observations into a mathematical equation called the Arrhenius equation (**Table 1**).

Table 1.The Arrhenius equation.

Equation	Symbols	Units
$k = Ae^{-\frac{E_a}{RT}}$	k = rate constant	various $\text{mol}^{-1}\text{s}^{-1}$
	A = the Arrhenius factor	various $\text{mol}^{-1}\text{s}^{-1}$
	T = temperature	kelvin (K)
	R = gas constant (the value of which is given in the DP chemistry data booklet as $8.31 \text{ J K}^{-1}\text{mol}^{-1}$)	joules per mole per kelvin ($\text{J K}^{-1}\text{mol}^{-1}$)
	E_a = activation energy	joules per mole (J mol^{-1})
	e = Euler's number This is the base of the natural logarithm (it has an approximate value of 2.718)	no units

The Arrhenius factor, A , is also called the pre-exponential factor. It takes into account the frequency and orientation of collisions. It varies slightly with temperature but not by much. It is generally considered constant across all temperatures and is specific for each reaction.

We know that for a successful collision to occur, particles must collide with:

- the proper orientation
- energy equal to or greater than the activation energy.

The Arrhenius factor takes into account the first point above. It tells us the frequency of collisions with the proper orientation, which is unique for every reaction.

It has the same units as k and can be deduced from the equation as follows:

$$A = \frac{k}{e^{-\frac{E_a}{RT}}}$$

The second point, the fraction of molecules with enough energy to react is represented by the next part of the equation that is:

$$e^{-\frac{E_a}{RT}}$$

This shows us that the fraction of molecules with enough energy depends on both temperature T , and the value of activation energy, E_a . The universal gas constant is a constant and does not impact the number of molecules with sufficient energy for a successful collision.

The Arrhenius equation can be rearranged mathematically, to determine the value of any of the variables in the equation depending on the given data:

- For any given reaction variables like temperature and rate (therefore rate constant) can be measured experimentally. R and e are constants.
- This leaves A and E_a , which must then be determined indirectly. Two unknowns require two equations, but as scientists we require several data points for validity, therefore graphing a series of data is required.
- The Arrhenius equation can be rearranged to form a linear equation. The independent and dependent variables can then be plotted as shown below. T is the independent variable as it is easily manipulated and k is the dependent variable, determined after rate is measured.

Graphical representation of the Arrhenius equation

1. Calculating the activation energy

Take the natural logarithm of both sides of the equation gives:

$$\begin{aligned}\ln k &= -\frac{E_a}{RT} + \ln A \\ &= -\left(\frac{E_a}{R}\right)\left(\frac{1}{T}\right) + \ln A\end{aligned}$$

This equation is in the form of a straight line equation $y = mx + c$.

Hence a graph of $\ln k$ (y-axis) against $\frac{1}{T}$ (x-axis) will be straight line with a gradient of $-\frac{E_a}{R}$.

This is known as the Arrhenius plot.

As R is a constant the activation energy can be calculated once the gradient of the graph is obtained:

$$\text{gradient} = -\frac{E_a}{R}$$

For example, if we were to measure the rate of degradation of a chemical at different temperatures, we could use these values to calculate the rate constant at those different temperatures from the rate expression $\text{rate} = k [\text{chemical}]$.

Plotting the $\ln k$ against $\frac{1}{T}$ will give the graph shown in **Figure 1**.

The gradient of the graph can be calculated and the value inserted into gradient $= -\frac{E_a}{R}$ to calculate E_a .

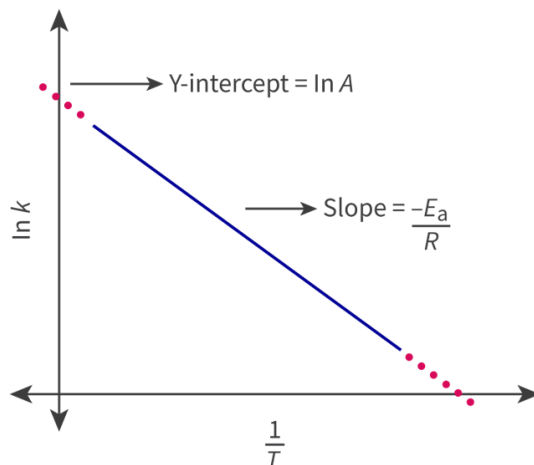


Figure 1. Arrhenius plot obtained by plotting experimentally obtained values of $\ln k$ vs $\frac{1}{T}$.

🔗 More information for figure 1

2. Calculating the Arrhenius factor

Once we have drawn the Arrhenius plot using the experimental data we can determine the value of the Arrhenius factor as it is given by the y- intercept on the graph in the logarithmic form of the Arrhenius equation:

$$\ln k = -\frac{E_a}{RT} + \ln A$$

The graph can be extrapolated to find the value of $\ln A$, as $\ln A$ is the y-intercept as shown in **Figure 1**. This value can be used to calculate A .

Alternatively, the Arrhenius equation can be rearranged to:

$$A = \frac{k}{e^{-\frac{E_a}{RT}}}$$

and A can be calculated.

Study skills

- The temperature, T , is taken in kelvin in the Arrhenius equation. In order to convert degrees Celsius to kelvin just add 273.15. That is, $K = ^\circ\text{C} + 273.15$
- The activation energy, E_a , is given in J mol^{-1} . To convert to kJ mol^{-1} , divide by 1000.

Worked example 1

The following data was collected for a reaction.

Rate constant, k , (s^{-1})	Temperature ($^\circ\text{C}$)
0.00049	392

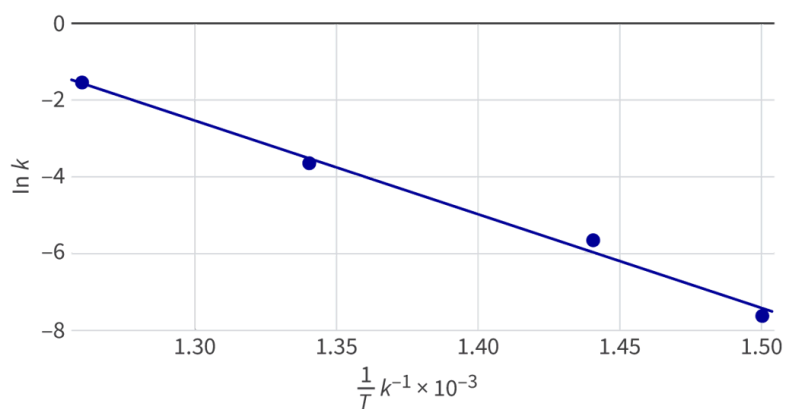
Rate constant, k , (s^{-1})	Temperature ($^{\circ}\text{C}$)
0.0035	421
0.026	475
0.21	521

1. Calculate the value of $\ln k$ and $\frac{1}{T}$
2. Plot the graph of $\ln k$ vs $\frac{1}{T}$
3. Measure the gradient of the line and calculate the activation energy in kJ mol^{-1} .

Convert the k values to $\ln k$ and the degrees Celsius to kelvin. Divide the kelvin values by 1 to get $\frac{1}{T}$.

$\ln k$	
-7.62	
-5.65	
-3.65	
-1.56	

Plotting the graph gives:



The gradient of the line is approximately -25000 K .

Using the equation;

$$\begin{aligned}\text{gradient} &= -\frac{E_a}{R} \\ -25000 &= -\frac{E_a}{8.31} \\ E_a &= (25000 \times 8.31) / 1000 \\ &= 208 \text{ kJ mol}^{-1}\end{aligned}$$

Please note that the final answer may change slightly depending on the value of the gradient.

Practical skills

Tool 1: Experimental techniques — Measuring variables, applying techniques

Tool 2: Technology — Applying technology to collect and manipulate data

Tool 3: Mathematics — Applying general mathematics and processing uncertainties

Inquiry 1: Exploring and designing — Controlling variables

Inquiry 2: Collecting and processing data — Interpreting results

Inquiry 3: Concluding and evaluating

A full investigation can be designed on determining the activation energy of a reaction using the Arrhenius equation, by varying the temperature of the reaction mixture.

Independent variable: Temperature

Dependent variable: k determined from the rate of reaction, calculated by measuring the change in concentration of the reactants over time at different temperatures.

Depending on the reaction under investigation, data loggers can be used to measure rate, followed by drawing an Arrhenius plot in Excel. Use the gradient of the plot to calculate E_a .

Calculation of activation energy without a plot

The activation energy can also be calculated using the values of the rate constant at only two different temperatures by using the equation:

$$\ln \left(\frac{k_1}{k_2} \right) = \frac{E_a}{R} \left(\frac{1}{T_2} - \frac{1}{T_1} \right)$$

Worked example 2

The following table shows data for the decomposition of hydrogen iodide at two different temperatures:

Rate Constant (k)	Temperature (T)
$k_1 = 7.78 \times 10^{-7}$	$T_1 = 273 \text{ K}$
$k_2 = 3.46 \times 10^{-5}$	$T_2 = 298 \text{ K}$

Calculate the activation energy for the reaction.

Rearranging and substituting the values into:

$$\ln \left(\frac{k_1}{k_2} \right) = \frac{E_a}{R} \left(\frac{1}{T_2} - \frac{1}{T_1} \right)$$

gives:

$$\begin{aligned}
 E_a &= \frac{R \ln \left(\frac{k_1}{k_2} \right)}{\left(\frac{1}{T_2} - \frac{1}{T_1} \right)} \\
 &= \frac{8.31 \ln \left(\frac{7.78 \times 10^{-7}}{3.46 \times 10^{-5}} \right)}{\left(\frac{1}{298} - \frac{1}{273} \right)} \\
 &= 1.03 \times 10^5 \text{ J mol}^{-1}
 \end{aligned}$$

Activity

- **IB learner profile attribute:**
 - Knowledgeable
 - Inquirer
 - Thinker
 - Reflective
- **Approaches to learning:**
 - Thinking skills — Providing a reasoned argument to support a conclusion, Applying key ideas and facts in new context
 - Communication skills — Using terminologies, symbols and communication conventions consistently and correctly
 - Social skills — Working collaboratively to achieve a common goal
- **Time required to complete activity:** 15—20 minutes
- **Activity type:** Pair activity

There is a familiar rule of thumb in chemistry which says that for reactions carried out at room temperature, for every 10 °C rise in temperature the rate of reaction doubles.

Perform the following activity in pairs and comment on the accuracy of this rule.

1. Use the Arrhenius equation to calculate the rate constants for a reaction where temperature is first 298 K and then it is increased to 308 K given the activation energy is taken as $E_a = 50 \text{ kJ mol}^{-1}$ at both temperatures. Assume that the Arrhenius factor is constant under all conditions.
2. Assuming the same 10 K rise in temperature, that is from 298 K to 308 K, calculate the new rate constants at 298 K and 308 K when the activation energy is 25 kJ mol^{-1} . Assume the Arrhenius factor is constant.
3. By what factor has the rate constant changed in both step 1 and step 2 (obtained by dividing the rate constant at 308 K with the rate constant at 298 K)?
4. Study the results obtained and brainstorm to comment on the reliability and accuracy of the above rule of thumb.

Doing a calculation and increasing temperature from to 298 K to 308 K and taking $E_a = 50 \text{ kJ mol}^{-1}$ ($48\,500 \text{ J mol}^{-1}$).

$$\begin{aligned} k_{298} &= Ae^{-\frac{E_a}{RT}} \\ &= Ae^{-\frac{50000}{8.31 \times 298}} \\ &= 1.70 \times 10^{-9} \text{ A} \end{aligned}$$

$$\begin{aligned} k_{308} &= Ae^{-\frac{E_a}{RT}} \\ &= Ae^{-\frac{50000}{8.31 \times 308}} \\ &= 3.28 \times 10^{-9} \text{ A} \end{aligned}$$

We can see that the rate constant has increased by a factor of $\frac{3.28 \times 10^{-9} \text{ A}}{1.70 \times 10^{-9} \text{ A}} = 1.9$ which is approximately twice the initial rate.

However, if we were to decrease the activation energy to 25 kJ mol^{-1} , substituting the value with the same rise in temperature gives:

$$\begin{aligned} k_{298} &= Ae^{-\frac{E_a}{RT}} \\ &= Ae^{-\frac{25000}{8.31 \times 298}} \\ &= 4.13 \times 10^{-5} \text{ A} \end{aligned}$$

$$\begin{aligned} k_{308} &= Ae^{-\frac{E_a}{RT}} \\ &= Ae^{-\frac{25000}{8.31 \times 308}} \\ &= 5.73 \times 10^{-5} \text{ A} \end{aligned}$$

For a 10 kelvin rise in temperature the rate constant increases by a factor of $\frac{5.73 \times 10^{-5} \text{ A}}{4.13 \times 10^{-5} \text{ A}} = 1.39$ which is approximately 1.4 times.

Activation energy values below and above 50 kJ mol^{-1} do not double the rate of reaction when temperature is increased by 10 K. The rule of thumb only applies to activation energies close to 50 kJ mol^{-1} and at temperatures close to room temperature.

The rule is not mathematically accurate.

